Disputes

Disputes raised against bids and payouts, with their evidence.

All 2 Open Escalated 1 Closed

D002 G002 · cycle 1 · bid
I believe a competing bid in this cycle was suspicious.
m4 · 28 Oct, 05:13 pm
Escalated

D001 G002 · cycle 1 · bid
I believe a competing bid in this cycle was suspicious.
m4 · 28 Oct, 05:13 pm
Resolved

Dispute D001

G002 · cycle 1 · against a bid · raised 28 Oct, 05:13 pm

Resolved

The complaint

I believe a competing bid in this cycle was suspicious.

Comments

[2026-10-28T11:44:27.067Z FOREMAN Rajesh (Foreman)] No, it's fine/

Evidence attached automatically

Captured 28 Oct, 05:13 pm when the dispute was raised.

Bids in this cycle

Member	Bid	Deviation	Status
m4	₹60,000	6.25σ	flagged suspicious
Member2	₹89,500	-1.13σ	accepted
m4	₹90,000	-1.25σ	accepted

Fraud-check logs

Rule	Details	Outcome
Underbid beyond the flag line	Abnormal underbid detected: Bid of ₹60,000 is 6.25 standard deviations below historical mean (₹85,000, σ=₹4,000). Threshold is 2.5σ.	Open

Ledger entries

#	Time	Action	Hash
23	28 Oct, 05:02 pm	contribution escrow locked	d5698067126e…
24	28 Oct, 05:04 pm	cycle opened	676e73268412…
25	28 Oct, 05:05 pm	cycle timer extended	d8899eece96c…
26	28 Oct, 05:05 pm	cycle timer extended	4267154ed591…
27	28 Oct, 05:05 pm	cycle timer extended	528f132e6689…
28	28 Oct, 05:07 pm	bid submitted	1e2f7b256615…
29	28 Oct, 05:08 pm	bid submitted	a5738c5eb4c8…
30	28 Oct, 05:10 pm	bid flagged fraud	93225d3fd598…

ChainChit | Dashboard | KYC | Groups | Bidding room • 1 live | Disputes

m4
Member · Trust 100

Sign out

Hello, m4

Your contributions, bids and payouts in one place.

Trust score	KYC	Contributed	Payouts won
100	• KYC verified	₹33,333	₹0
Out of 100		1 payments	0 cycles

My groups

Silver Wealth Chit G002

₹1,00,000 pool · ₹33,333 per cycle

Active

Contributions to pay

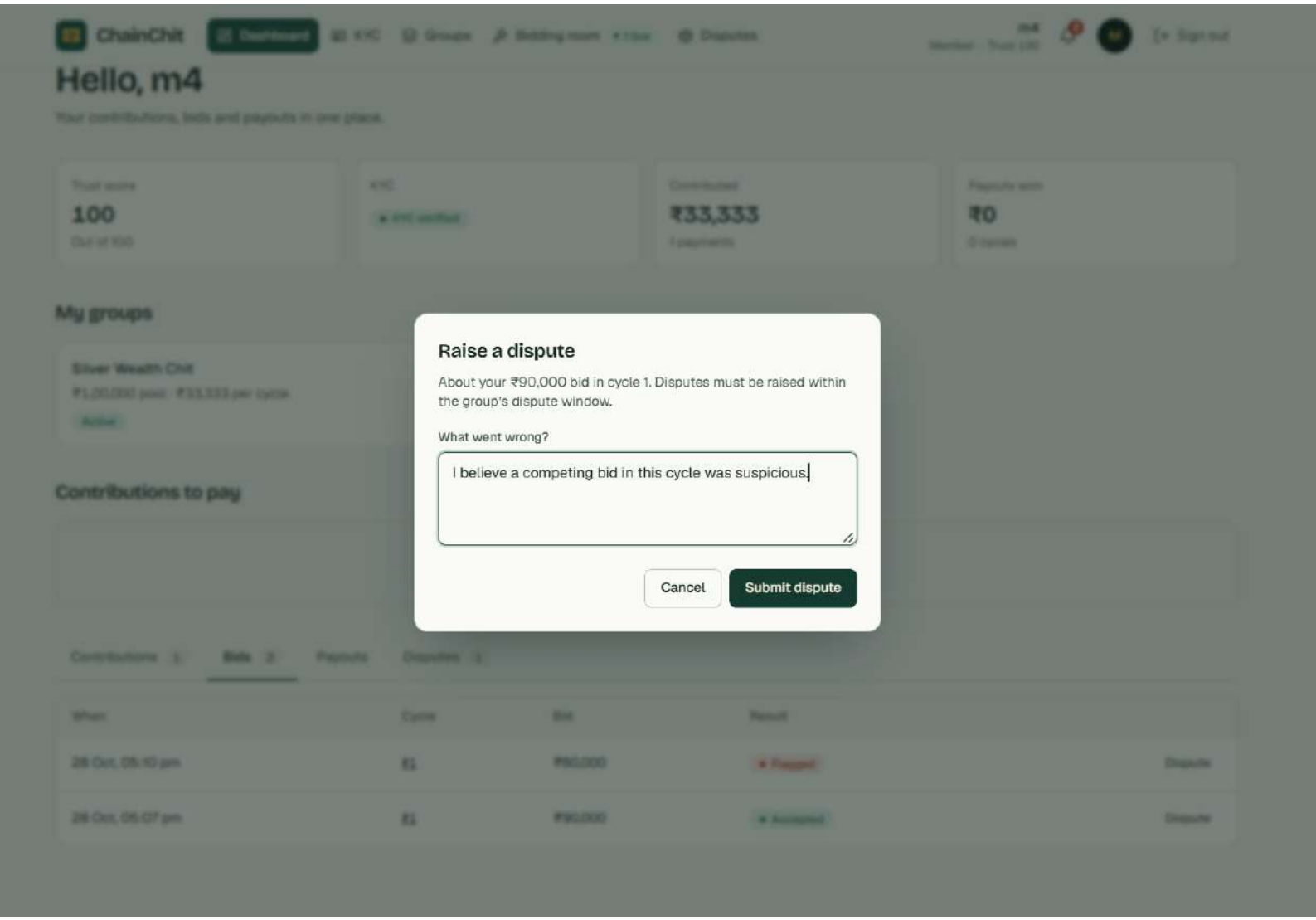
Nothing to pay right now.

Contributions 1 | **Bids 2** | Payouts | Disputes 2

When	Cycle	Bid	Result	
28 Oct, 05:10 pm	#1	₹60,000	• Flagged	Dispute
28 Oct, 05:07 pm	#1	₹90,000	• Accepted	Dispute

Dispute D002 raised

The foreman and oversight viewer have been told. Your transaction history and fraud-check logs were attached.


Raise a dispute
About your ₹90,000 bid in cycle 1. Disputes must be raised within the group's dispute window.
What went wrong?
I believe a competing bid in this cycle was suspicious.
Cancel
Submit dispute

ChainChit | Dashboard | Groups | Bidding room | 1 live | Foreman desk | Disputes | Audit ledger | Rajesh (Foreman) Foreman | Sign out

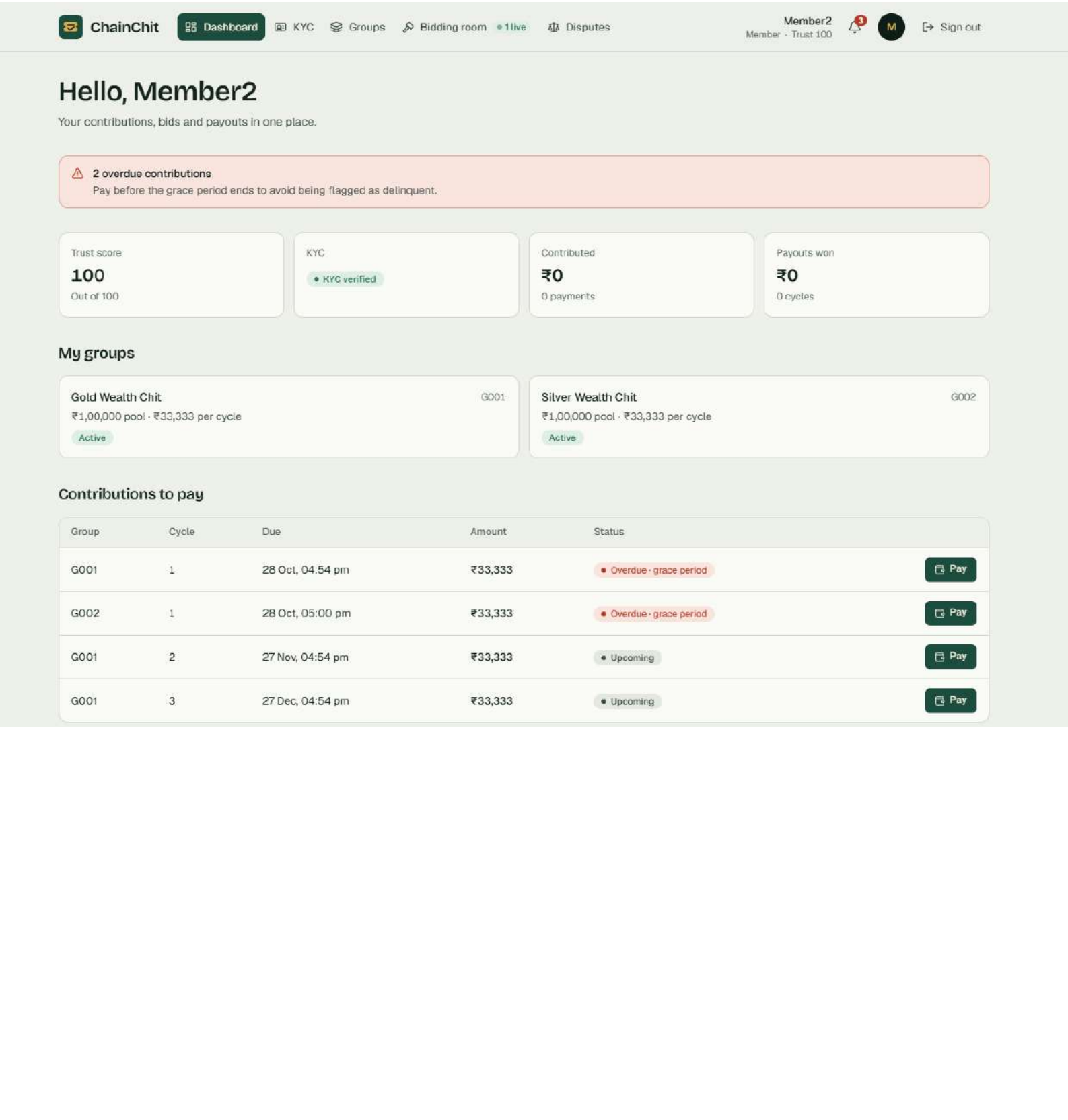
Audit ledger

Every bid, ruling and payout is written here and chained to the entry before it. If anyone edits an old entry, the chain breaks and the check below says so.

Check chain | Export signed PDF | Download CSV

Chain intact
All 30 ledger entries verified: SHA-256 chain intact, Ed25519 signatures valid. (30 entries checked)

When	Who	What happened	On	Fingerprint
28 Oct, 05:10 pm	m4 Member	Bid flagged fraud	bid	93225d3fd5…a5c9
28 Oct, 05:08 pm	Member2 Member	Bid submitted	bid	a5738c5eb4…3213
28 Oct, 05:07 pm	m4 Member	Bid submitted	bid	1e2f7b2566…a75d
28 Oct, 05:05 pm	Rajesh (Foreman) Foreman	Cycle timer extended	cycle	528f132e66…5e72
28 Oct, 05:05 pm	Rajesh (Foreman) Foreman	Cycle timer extended	cycle	4267154ed5…b7e7
28 Oct, 05:05 pm	Rajesh (Foreman) Foreman	Cycle timer extended	cycle	d8899ecce9…e147
28 Oct, 05:04 pm	Rajesh (Foreman) Foreman	Cycle opened	cycle	676a732684…03e3
28 Oct, 05:02 pm	m4 Member	Contribution escrow locked	contribution	d569806712…a750
28 Oct, 05:02 pm	Member1 Member	Contribution escrow locked	contribution	129494d99f…234b
28 Oct, 05:00 pm	Rajesh (Foreman) Foreman	Group created	chit group	0ee0f5f0dd…c235
28 Oct, 04:57 pm	Rajesh (Foreman) Foreman	Kyc document viewed	kyc document	f485ab7358…f8c3
28 Oct, 04:57 pm	Rajesh (Foreman) Foreman	Kyc verified	user	46f42a848c…a125


ChainChit
Dashboard
KYC
Groups
Bidding room
1 live
Disputes
Member2
Member · Trust 100
3
M
Sign out
Hello, Member2
Your contributions, bids and payouts in one place.
2 overdue contributions
Pay before the grace period ends to avoid being flagged as delinquent.
Trust score
100
Out of 100
KYC
KYC verified
Contributed
₹0
0 payments
Payouts won
₹0
0 cycles
My groups
Gold Wealth Chit
G001
₹1,00,000 pool · ₹33,333 per cycle
Active
Silver Wealth Chit
G002
₹1,00,000 pool · ₹33,333 per cycle
Active
Contributions to pay
Group
Cycle
Due
Amount
Status
G001
1
28 Oct, 04:54 pm
₹33,333
Overdue · grace period
Pay
G002
1
28 Oct, 05:00 pm
₹33,333
Overdue · grace period
Pay
G001
2
27 Nov, 04:54 pm
₹33,333
Upcoming
Pay
G001
3
27 Dec, 04:54 pm
₹33,333
Upcoming
Pay

Notifications

Reminders, KYC decisions, fraud flags, payouts and dispute updates.

Mark all read

- **Bid flagged** in G002: m4 bid ₹60,000. Abnormal underbid detected: Bid of ₹60,000 is 6.25 standard deviations below historical mean (₹85,000, σ=₹4,000). Threshold is 2.5σ.
  28 Oct, 05:10 pm

Notifications

Reminders, KYC decisions, fraud flags, payouts and dispute updates.

Mark all read

- Your ₹33,333.33 contribution for G002 cycle 1 is overdue. Pay within the grace period to avoid being flagged.
  28 Oct, 05:01 pm
- Your ₹33,333.33 contribution for G001 cycle 1 is overdue. Pay within the grace period to avoid being flagged.
  28 Oct, 05:01 pm
- Your KYC has been verified.
  28 Oct, 03:09 pm

ChainChit | Dashboard | KYC | Groups | Bidding room • 1 live | Disputes

m4
Member · Trust 100

Sign out

Silver Wealth Chit

Cycle 1 of 3 • Bidding open

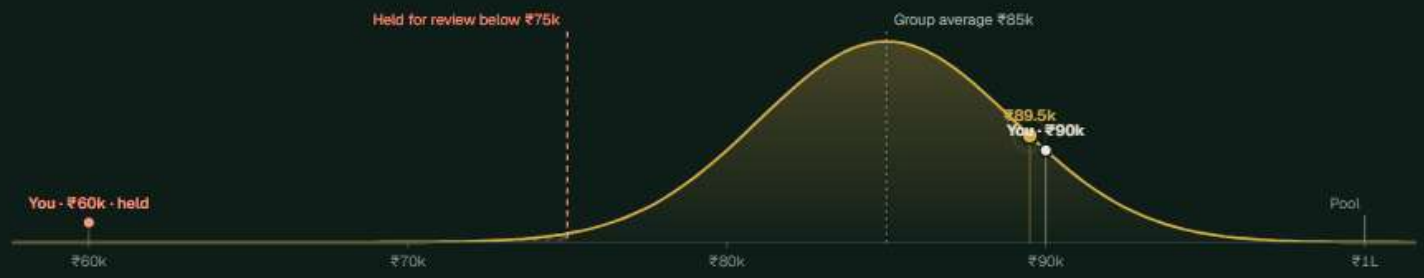
Window closes in
05:50

Lowest valid bid

₹89,500

Member2 is ahead - ₹10,500 below the pool

Pool	Members enrolled	Bids	Held by fraud check
₹1,00,000	3 of 3	3	ⓘ 1



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

3 bids

You 05:10:30 pm Abnormal underbid detected: Bid of ₹60,000 is 6.25 standard deviations below historical mean (₹85,000, σ=₹4,000). Threshold is 2.5σ.	ⓘ Held	~~₹60,000~~
Member2 05:08:06 pm	Leading	₹89,500
You 05:07:14 pm		₹90,000

Your bid

Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

Bid amount

₹ 89,000

Beat the lead by -500 -1,000 -2,500

Pool value	₹1,00,000
Prize you take	—
Discount shared with the group	—
Dividend per member (÷ 3)	—
Your monthly due after dividend	—

ⓘ **₹60,000 is held for review**
Abnormal underbid detected: Bid of ₹60,000 is 6.25 standard deviations below historical mean (₹85,000, σ=₹4,000). Threshold is 2.5σ.

Place bid

Notifications

Reminders, KYC decisions, fraud flags, payouts and dispute updates.

Nothing yet

Updates about your contributions, bids and disputes will appear here.

ChainChit | Dashboard | KYC | Groups | Bidding room • 1 live | Disputes | Member2 Member · Trust 100 | 3 | M | Sign out

← All bidding rooms

• Live

Silver Wealth Chit

Cycle 1 of 3 • Bidding open

Window closes in

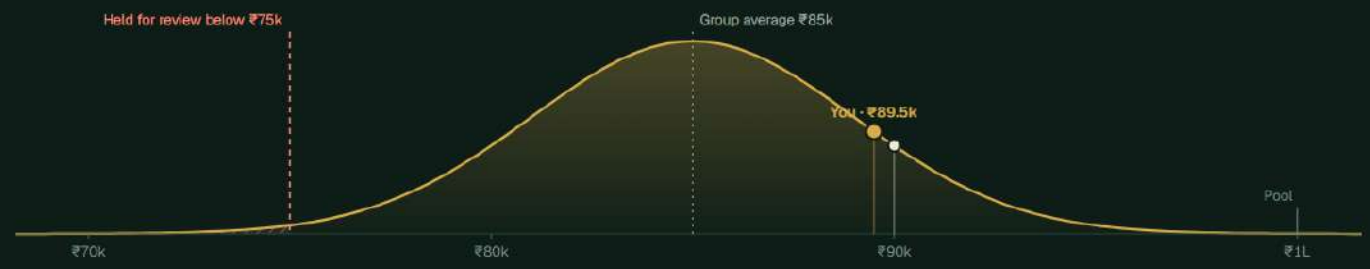
07:43

Lowest valid bid

₹89,500

You are ahead · ₹10,500 below the pool

Pool	Members enrolled	Bids
₹1,00,000	3 of 3	2



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

2 bids

You	05:08:06 pm	Leading	₹89,500

Your bid

Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

ChainChit | Dashboard | KYC | Groups | Bidding room · 1 live | Disputes

Member2
Member · Trust 100

Sign out

← All bidding rooms

● Live

Silver Wealth Chit

Cycle 1 of 3 · Bidding open

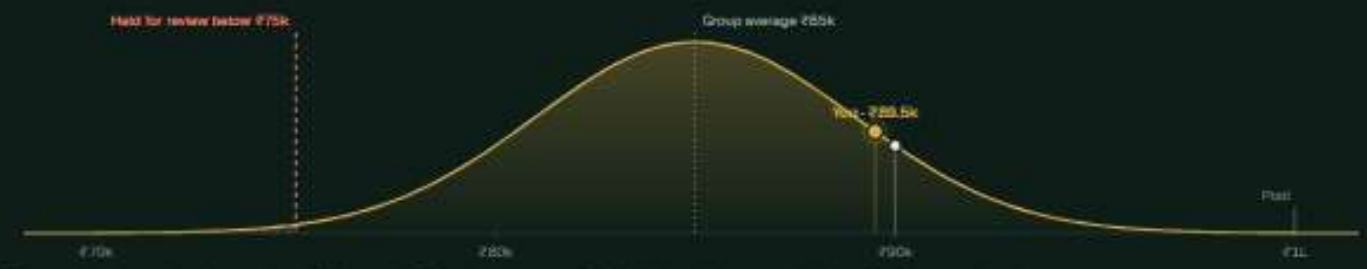
Window closes in
08:07

Lowest valid bid

₹89,500

You are ahead · ₹10,500 below the pool

Pool	Members enrolled	Bids
₹1,00,000	3 of 3	2



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

2 bids

You 05:08:06 pm	Leading	₹89,500
m4 05:07:14 pm		₹90,000

Your bid

Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

Bid amount

₹ 89,000

Beat the lead by -500 -1,000 -2,500

Pool value	₹1,00,000
Prize you take	—
Discount shared with the group	—
Dividend per member (÷ 3)	—
Your monthly due after dividend	—

You hold the lowest valid bid
You're at ₹89,500. You can bid again once another member goes lower.

Bid placed at ₹89,500.

Place bid

ChainChit | Dashboard | KYC | Groups | Bidding room • 1 live | Disputes | m4 Member · Trust 100 | Sign out

₹90,000

You are ahead - ₹10,000 below the pool

Pool	Members enrolled	Bids
₹1,00,000	3 of 3	1



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

1 bid

You	05:07:14 pm	Leading	₹90,000

Your bid

Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

Bid amount

₹ 89,500

Beat the lead by -500 -1,000 -2,500

Pool value	₹1,00,000
Prize you take	—
Discount shared with the group	—
Dividend per member (÷ 3)	—
Your monthly due after dividend	—

You hold the lowest valid bid
You're at ₹90,000. You can bid again once another member goes lower.

Bid placed at ₹90,000.

Place bid

ChainChit | Dashboard | Groups | Bidding room · 1 live | Foreman desk | Disputes | Audit ledger | Rajesh (Foreman) Foreman | Sign out

← All bidding rooms

● Live

Silver Wealth Chit

Cycle 1 of 3 · Bidding open

Window closes in

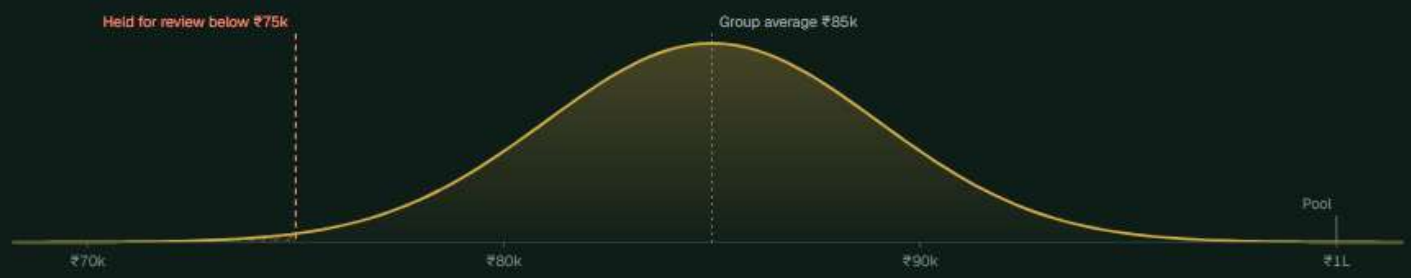
10:07

Lowest valid bid

No bid yet

The pool is ₹1,00,000. The first valid bid sets the pace and every bid after it must go lower to lead.

Pool	Members enrolled	Bids
₹1,00,000	3 of 3	0



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

0 bids

Nothing yet. Bids appear here the moment they are placed, lowest first.

Foreman controls

Escrow locked

Bidding ends on its own when the clock runs out. Close it now to name the winner early.

Close bidding now

Held for review

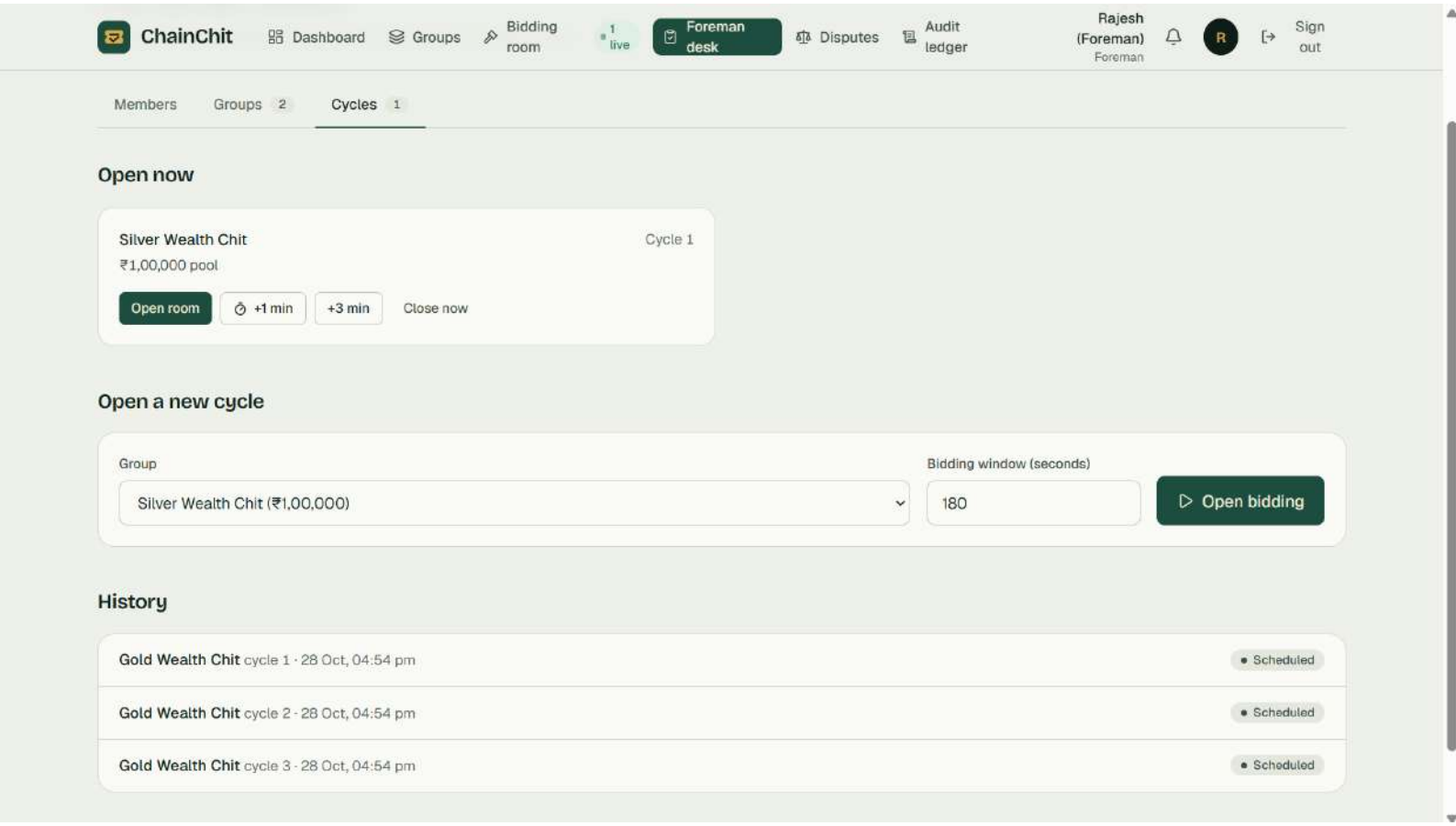
No bids are waiting on a ruling.

Demo scenarios

Your bid

Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

Foremen run the auction, they don't bid
Sign in as an enrolled, KYC-verified member to place bids.


ChainChit
Dashboard
Groups
Bidding room
1 live
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
R
Sign out
Members
Groups 2
Cycles 1
Open now
Silver Wealth Chit
Cycle 1
₹1,00,000 pool
Open room
+1 min
+3 min
Close now
Open a new cycle
Group
Silver Wealth Chit (₹1,00,000)
Bidding window (seconds)
180
Open bidding
History
Gold Wealth Chit cycle 1 · 28 Oct, 04:54 pm
Scheduled
Gold Wealth Chit cycle 2 · 28 Oct, 04:54 pm
Scheduled
Gold Wealth Chit cycle 3 · 28 Oct, 04:54 pm
Scheduled

Silver Wealth Chit

G002 · ₹1,00,000 pool · — per member each 30 days

Status	Members	Cycles settled	Fraud flag line
Active	3/3 Minimum 3	0/1	2.5σ Below the group's average bid

Schedule 1 Members 3 Ledger

Ledger verified
All 23 ledger entries verified: SHA-256 chain intact, Ed25519 signatures valid.

#	Time	Action	Hash
23	28 Oct, 05:02 pm	contribution escrow locked	d5698067126e...
21	28 Oct, 05:00 pm	group created	0ee0f5f0ddfd...



Chit groups

Every group, its schedule and its ledger.

Active 2 | Completed

G002 — Active

Silver Wealth Chit

₹1,00,000 pool · ₹33,333 each

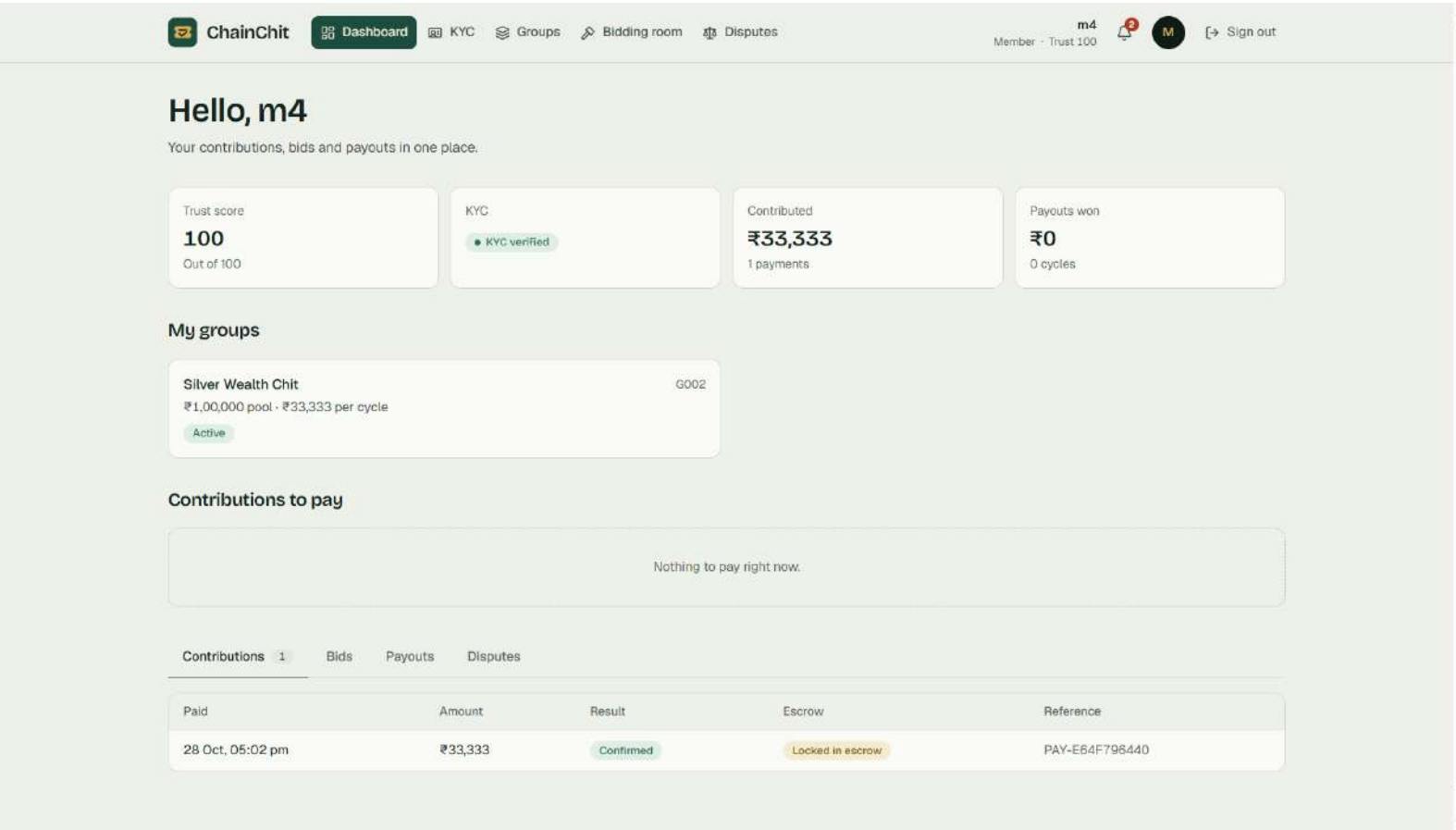
3/3 members · 0/1 cycles settled

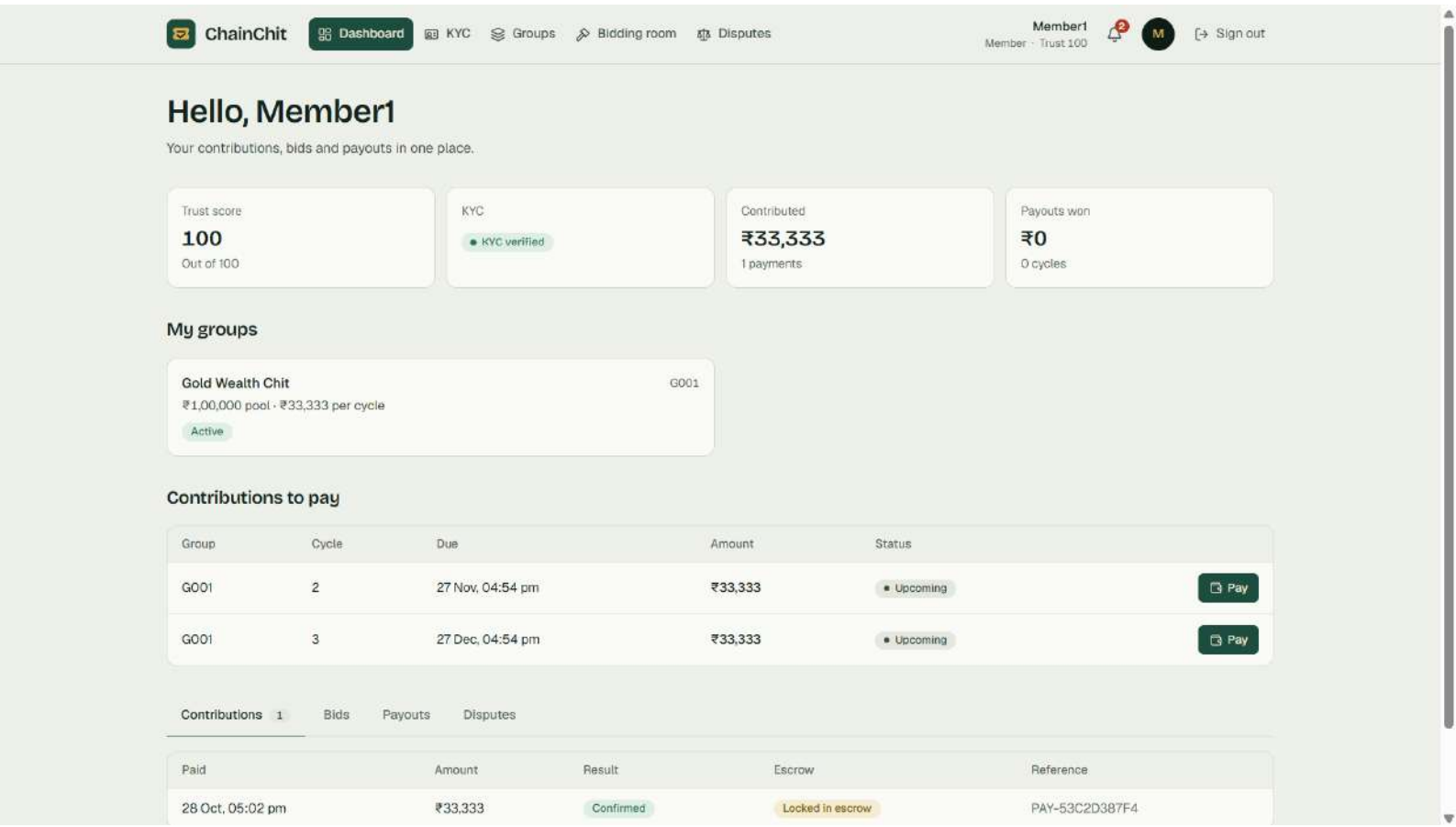
G001 — Active

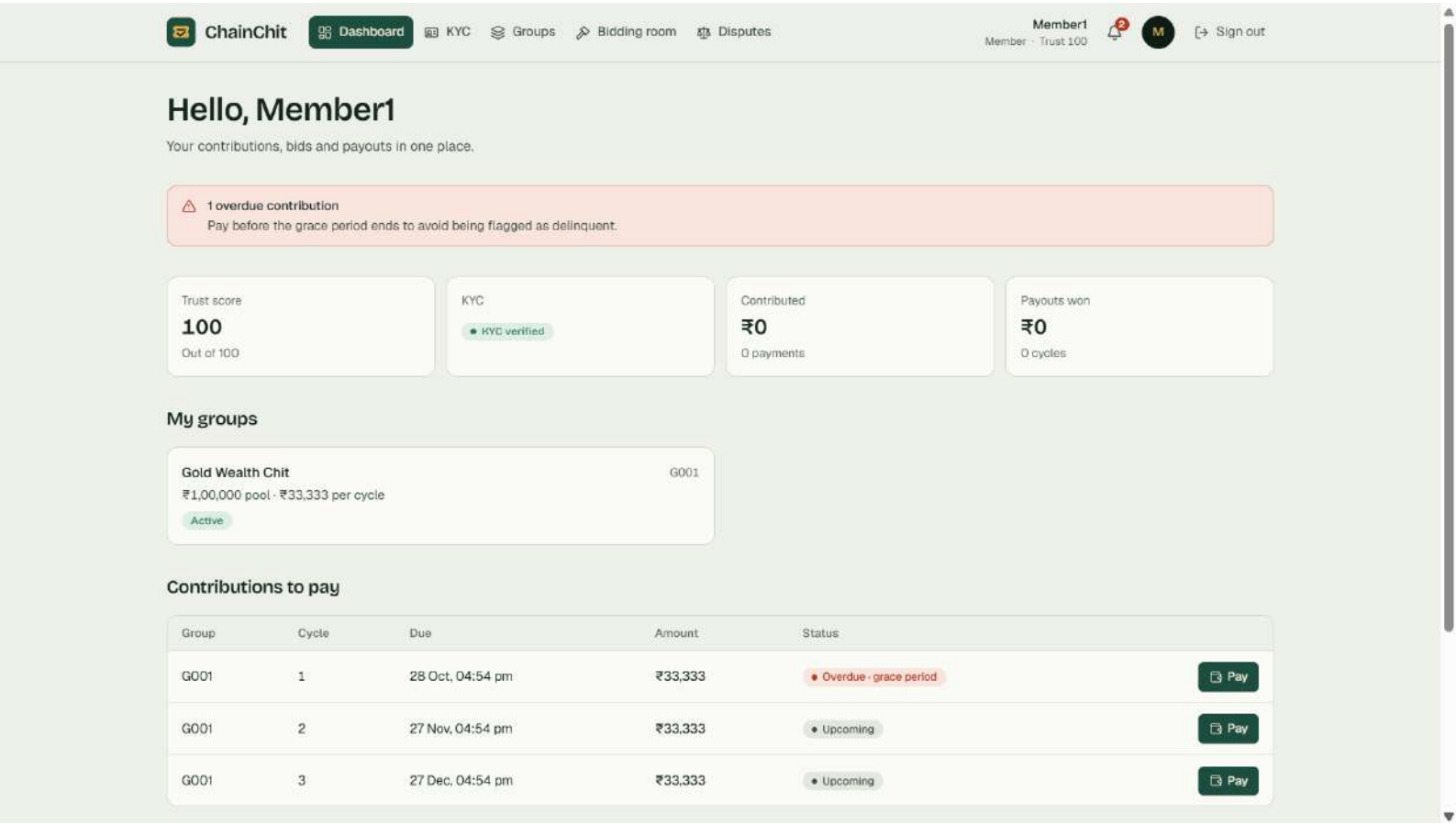
Gold Wealth Chit

₹1,00,000 pool · ₹33,333 each

3/3 members · 0/3 cycles settled


ChainChit
Dashboard
KYC
Groups
Bidding room
Disputes
m4
Member · Trust 100
2
M
Sign out
Hello, m4
Your contributions, bids and payouts in one place.
Trust score
100
Out of 100
KYC
KYC verified
Contributed
₹33,333
1 payments
Payouts won
₹0
0 cycles
My groups
Silver Wealth Chit
G002
₹1,00,000 pool · ₹33,333 per cycle
Active
Contributions to pay
Nothing to pay right now.
Contributions 1
Bids
Payouts
Disputes
Paid
Amount
Result
Escrow
Reference
28 Oct, 05:02 pm
₹33,333
Confirmed
Locked in escrow
PAY-E64F796440


ChainChit
Dashboard
KYC
Groups
Bidding room
Disputes
Member1
Member · Trust 100
2
M
Sign out
Hello, Member1
Your contributions, bids and payouts in one place.
Trust score
100
Out of 100
KYC
KYC verified
Contributed
₹33,333
1 payments
Payouts won
₹0
0 cycles
My groups
Gold Wealth Chit
G001
₹1,00,000 pool · ₹33,333 per cycle
Active
Contributions to pay
Group
Cycle
Due
Amount
Status
G001
2
27 Nov, 04:54 pm
₹33,333
Upcoming
Pay
G001
3
27 Dec, 04:54 pm
₹33,333
Upcoming
Pay
Contributions 1
Bids
Payouts
Disputes
Paid
Amount
Result
Escrow
Reference
28 Oct, 05:02 pm
₹33,333
Confirmed
Locked in escrow
PAY-53C2D387F4


ChainChit
Dashboard
KYC
Groups
Bidding room
Disputes
Member1
Member · Trust 100
2
M
Sign out
Hello, Member1
Your contributions, bids and payouts in one place.
1 overdue contribution
Pay before the grace period ends to avoid being flagged as delinquent.
Trust score
100
Out of 100
KYC
KYC verified
Contributed
₹0
0 payments
Payouts won
₹0
0 cycles
My groups
Gold Wealth Chit
G001
₹1,00,000 pool · ₹33,333 per cycle
Active
Contributions to pay
Group
Cycle
Due
Amount
Status
G001
1
28 Oct, 04:54 pm
₹33,333
Overdue · grace period
Pay
G001
2
27 Nov, 04:54 pm
₹33,333
Upcoming
Pay
G001
3
27 Dec, 04:54 pm
₹33,333
Upcoming
Pay

ChainChit | Dashboard | Groups | Bidding room | Foreman desk | Disputes | Audit ledger | Rajesh (Foreman) Foreman | R | Sign out



The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.

Bids so far

0 bids

Nothing yet. Bids appear here the moment they are placed, lowest first.

Foreman controls

Escrow locked

Held for review

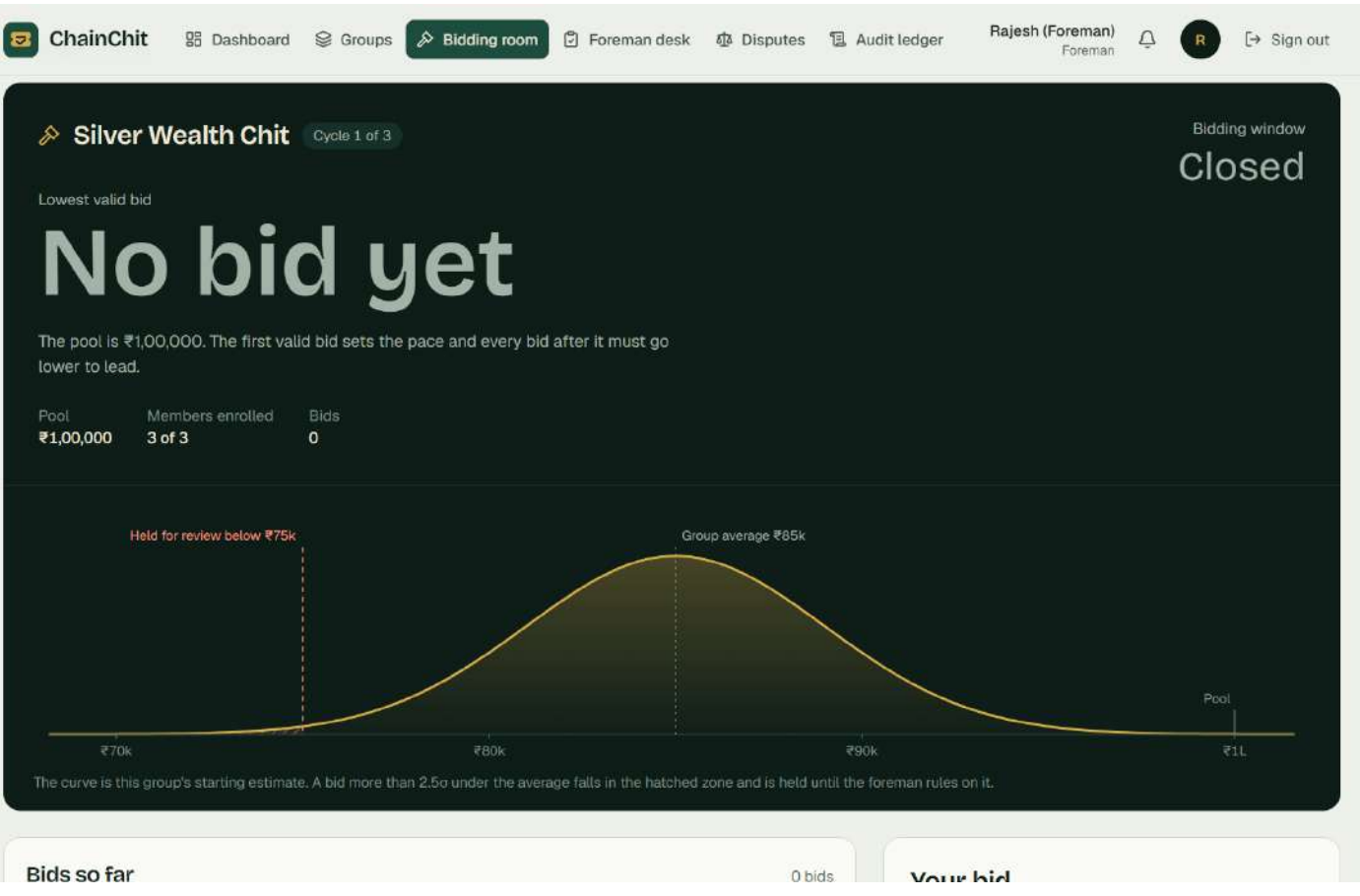
No bids are waiting on a ruling.

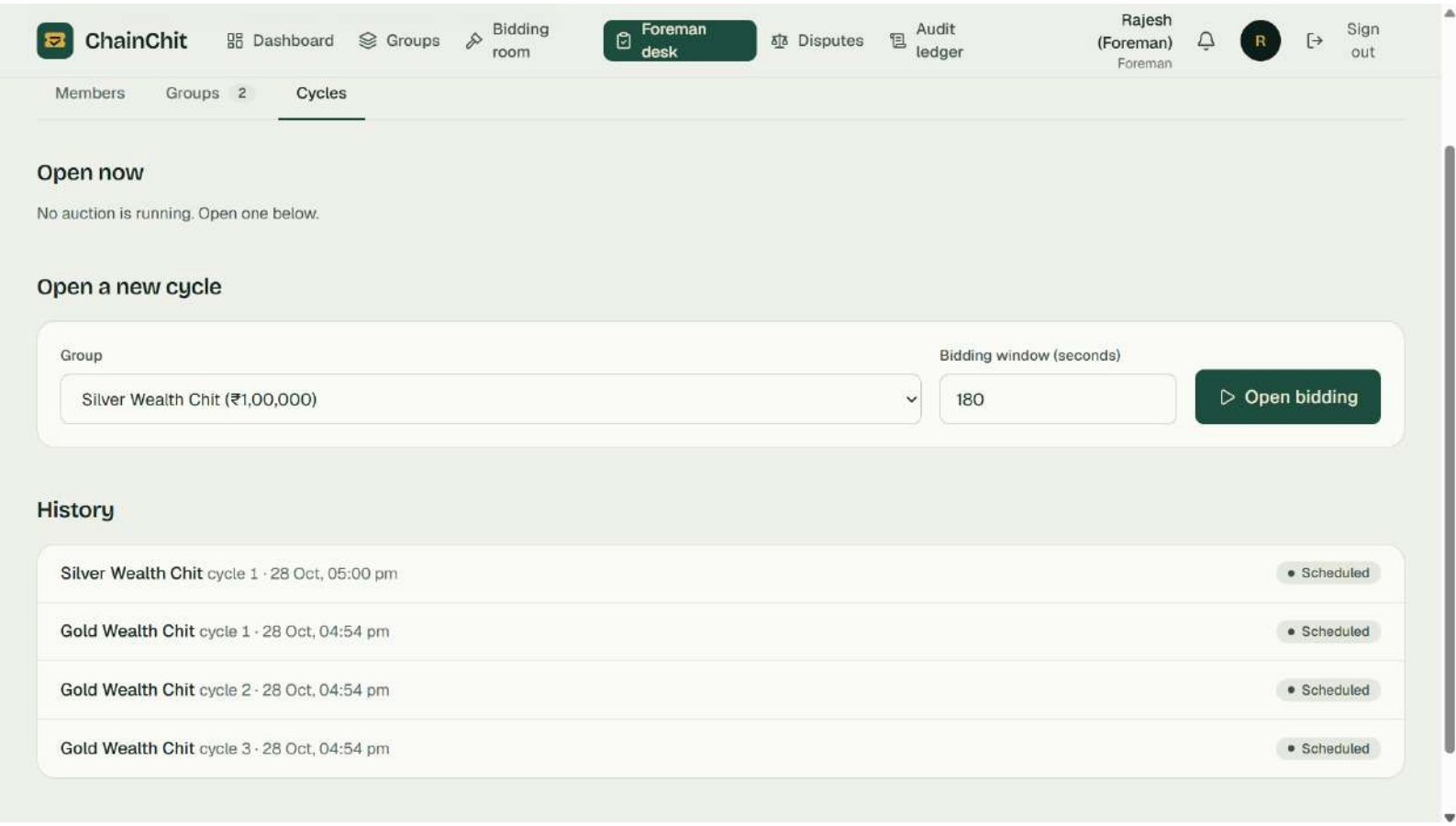
Demo scenarios

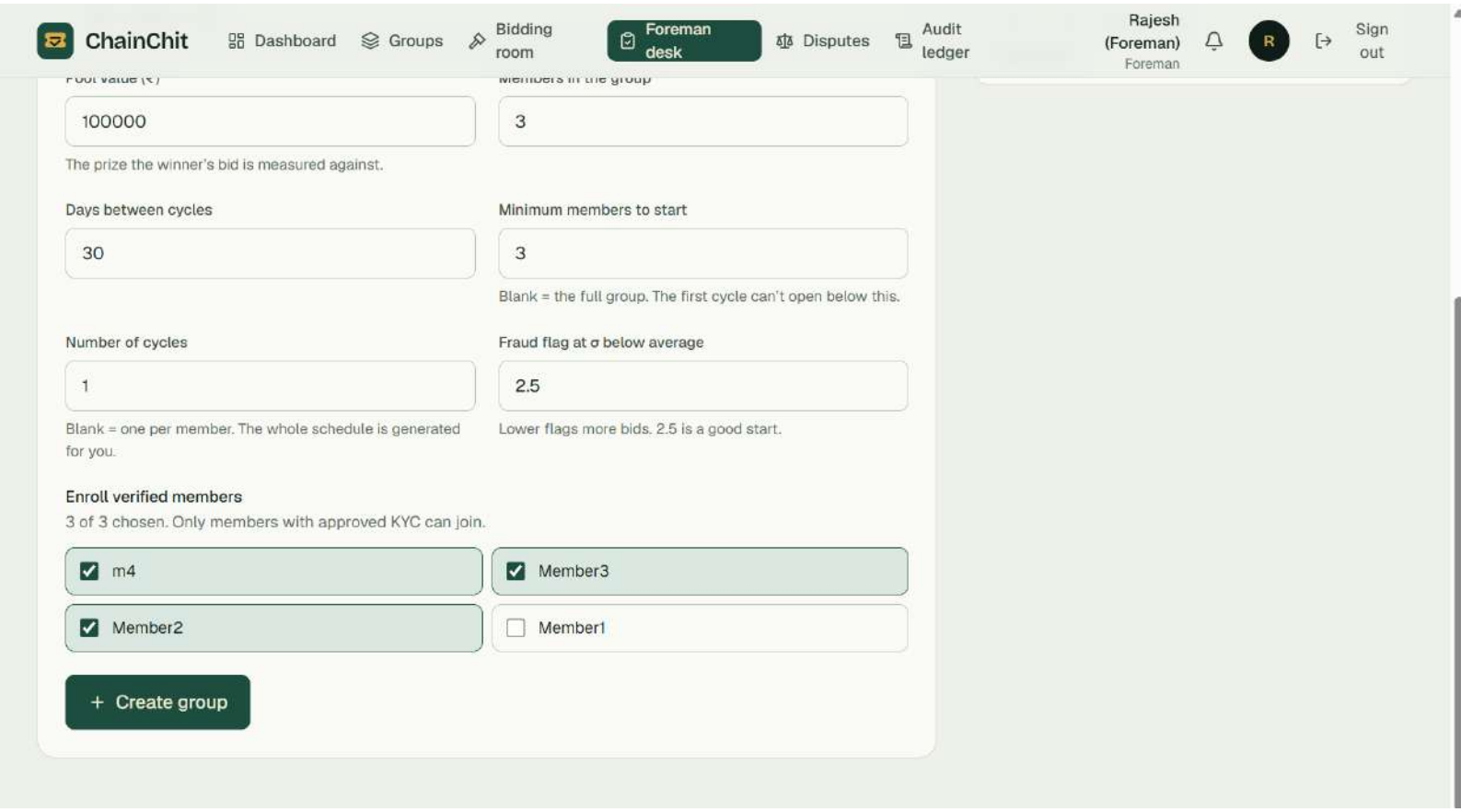
Your bid

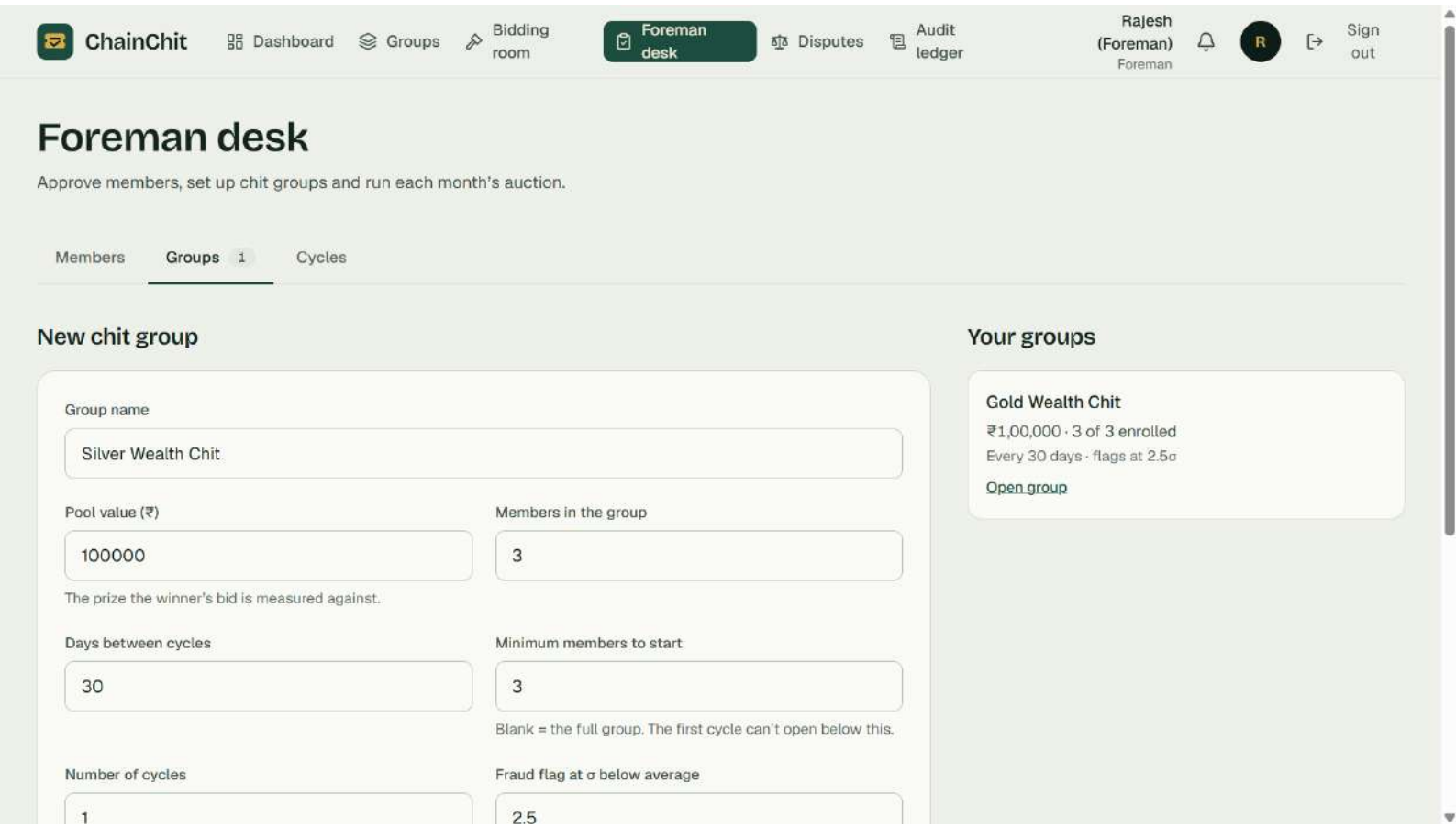
Name the lowest prize you'd accept from the ₹1,00,000 pool. The lowest valid bid wins.

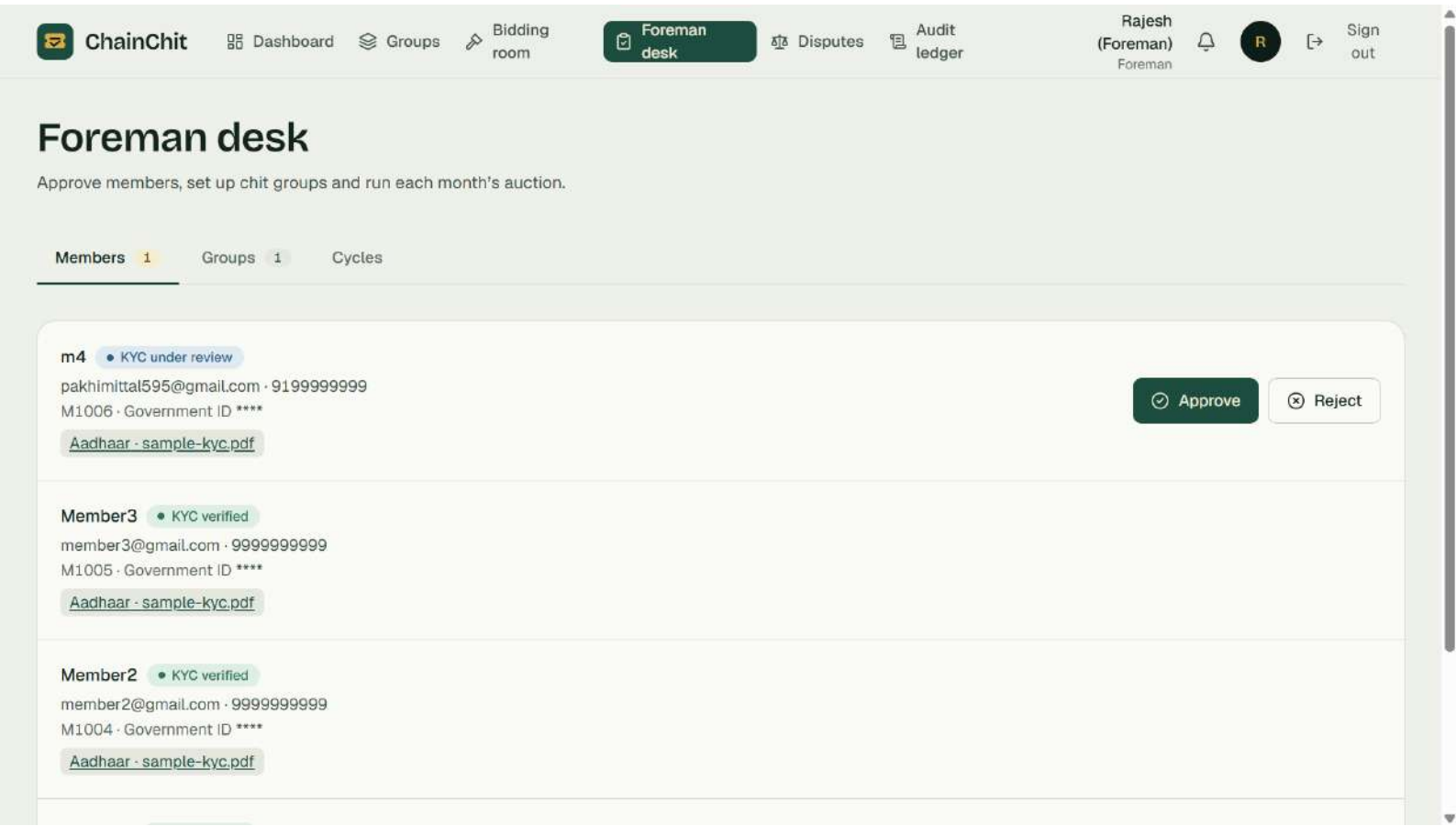
Bidding is closed for this cycle
The foreman hasn't opened the window yet.

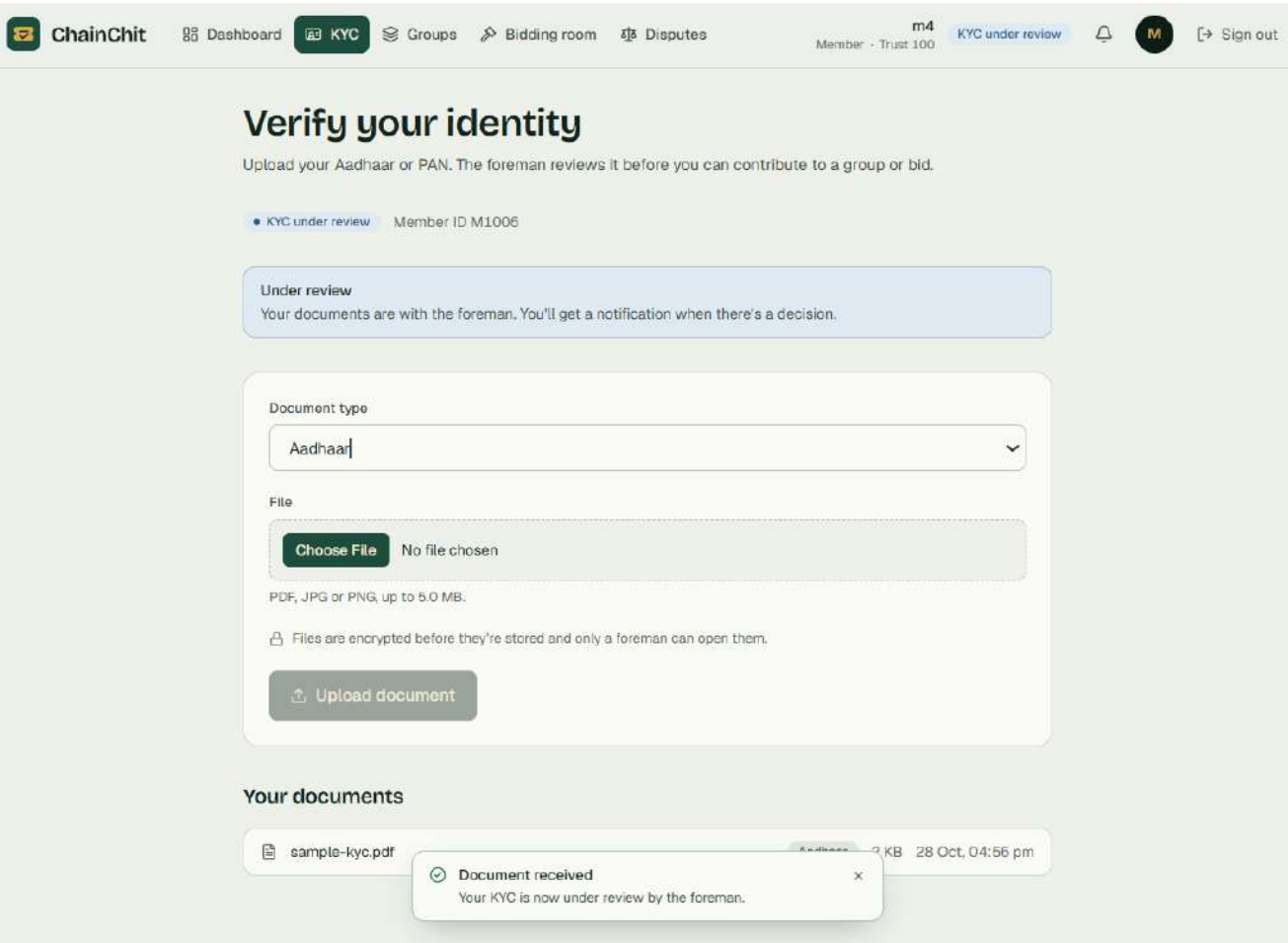

ChainChit
Dashboard
Groups
Bidding room
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
R
Sign out
Silver Wealth Chit
Cycle 1 of 3
Bidding window
Closed
Lowest valid bid
No bid yet
The pool is ₹1,00,000. The first valid bid sets the pace and every bid after it must go lower to lead.
Pool
₹1,00,000
Members enrolled
3 of 3
Bids
0
Held for review below ₹75k
Group average ₹85k
Pool
₹70k
₹80k
₹90k
₹1L
The curve is this group's starting estimate. A bid more than 2.5σ under the average falls in the hatched zone and is held until the foreman rules on it.
Bids so far
0 bids
Your bid


ChainChit
Dashboard
Groups
Bidding room
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
R
Sign out
Members
Groups 2
Cycles
Open now
No auction is running. Open one below.
Open a new cycle
Group
Silver Wealth Chit (₹1,00,000)
Bidding window (seconds)
180
Open bidding
History
Silver Wealth Chit cycle 1 · 28 Oct, 05:00 pm
Scheduled
Gold Wealth Chit cycle 1 · 28 Oct, 04:54 pm
Scheduled
Gold Wealth Chit cycle 2 · 28 Oct, 04:54 pm
Scheduled
Gold Wealth Chit cycle 3 · 28 Oct, 04:54 pm
Scheduled


ChainChit
Dashboard
Groups
Bidding room
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
Sign out
100000
3
The prize the winner's bid is measured against.
Days between cycles
30
Minimum members to start
3
Blank = the full group. The first cycle can't open below this.
Number of cycles
1
Fraud flag at σ below average
2.5
Blank = one per member. The whole schedule is generated for you.
Lower flags more bids. 2.5 is a good start.
Enroll verified members
3 of 3 chosen. Only members with approved KYC can join.
m4
Member3
Member2
Member1
+ Create group


ChainChit
Dashboard
Groups
Bidding room
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
R
Sign out
Foreman desk
Approve members, set up chit groups and run each month's auction.
Members
Groups 1
Cycles
New chit group
Group name
Silver Wealth Chit
Pool value (₹)
100000
Members in the group
3
The prize the winner's bid is measured against.
Days between cycles
30
Minimum members to start
3
Blank = the full group. The first cycle can't open below this.
Number of cycles
1
Fraud flag at σ below average
2.5
Your groups
Gold Wealth Chit
₹1,00,000 · 3 of 3 enrolled
Every 30 days · flags at 2.5σ
Open group


ChainChit
Dashboard
Groups
Bidding room
Foreman desk
Disputes
Audit ledger
Rajesh (Foreman)
Foreman
R
Sign out
Foreman desk
Approve members, set up chit groups and run each month's auction.
Members 1
Groups 1
Cycles
m4
KYC under review
pakhimittal595@gmail.com · 9199999999
M1006 · Government ID ****
Aadhaar · sample-kyc.pdf
Approve
Reject
Member3
KYC verified
member3@gmail.com · 9999999999
M1005 · Government ID ****
Aadhaar · sample-kyc.pdf
Member2
KYC verified
member2@gmail.com · 9999999999
M1004 · Government ID ****
Aadhaar · sample-kyc.pdf


ChainChit
Dashboard
KYC
Groups
Bidding room
Disputes
m4
Member · Trust 100
KYC under review
M
Sign out
Verify your identity
Upload your Aadhaar or PAN. The foreman reviews it before you can contribute to a group or bid.
KYC under review
Member ID M1006
Under review
Your documents are with the foreman. You'll get a notification when there's a decision.
Document type
Aadhaar
File
Choose File
No file chosen
PDF, JPG or PNG, up to 5.0 MB.
Files are encrypted before they're stored and only a foreman can open them.
Upload document
Your documents
sample-kyc.pdf
2 KB
28 Oct, 04:56 pm
Document received
Your KYC is now under review by the foreman.

ChainChit

Chit auctions you can audit.

Members bid the prize down in real time. Every bid is checked against the group's own history and every action is sealed into a SHA-256 chain that anyone with oversight can verify.

1. **Bid the prize down**
   Each member names the lowest payout they will accept from the pool.
2. **The lowest valid bid wins**
   A Z-score check holds any underbid that is statistically out of line.
3. **Everyone shares the discount**
   What the winner gives up is paid back to the group as dividends.

Permissioned groups · KYC-verified members · Escrow held until fraud checks clear

Create your account

Members are approved by the foreman before they can bid.

Full legal name

As on your ID

Email

Password

At least 6 characters.

Mobile number

PAN or Aadhaar

Your account starts as **KYC pending**. You can look around straight away; bidding opens once the foreman approves your ID.

Create account

Already registered? Sign in

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Permissioned groups · KYC-verified members · Escrow held until fraud checks clear

Sign in

Use the email and password you registered with.

Email

name@example.com

Password

Sign in

New to ChainChit? Create a member account

Trying the seeded demo?
Fill in the foreman login

```
[✓] Pulling schema from database...
[✓] Changes applied
triggers installed
Seeded: foreman@chainchit.io and oversight@chainchit.io (password from SEED_PASSWORD, default Password@123).
Members register themselves via /register.
PS C:\Users\pakhi\Downloads\chainchit-main> npm run dev

> chainchit@0.1.0 dev
```

Sign in

Use the email and password you registered with.

Email

oversight@chainchit.io

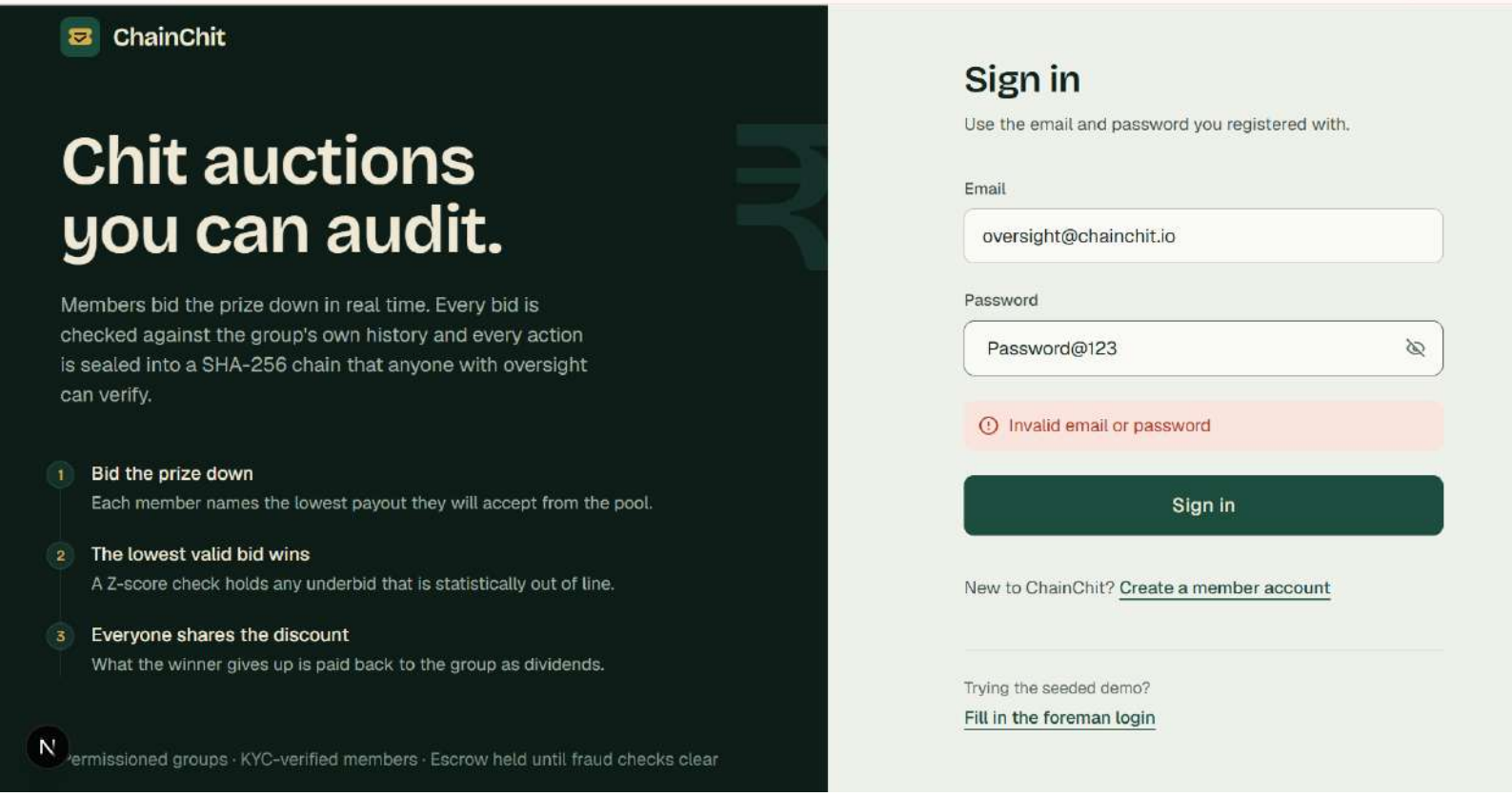
Password

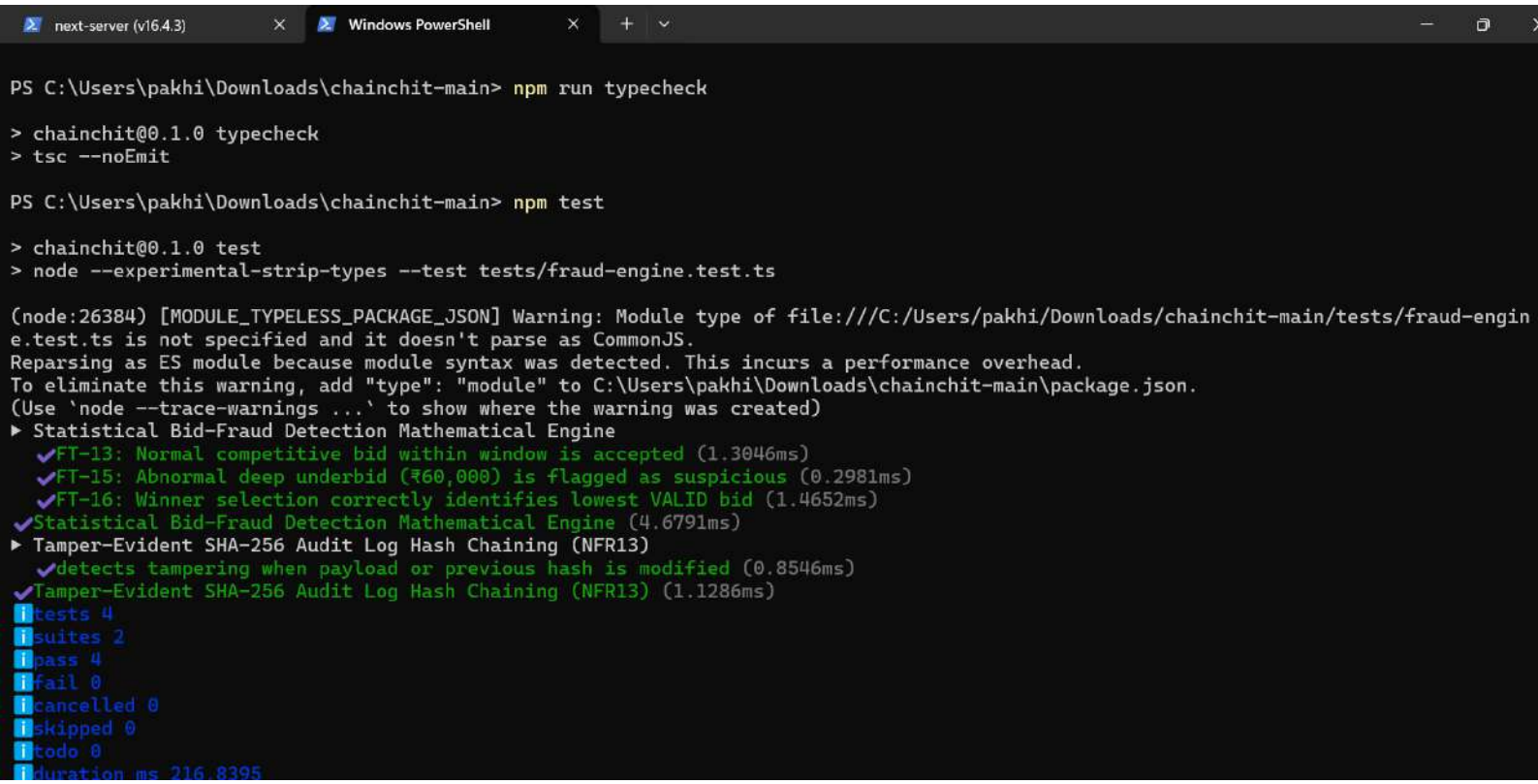
Password@123

Invalid email or password

Sign in

New to ChainChit? Create a member account


ChainChit
Chit auctions you can audit.
Members bid the prize down in real time. Every bid is checked against the group's own history and every action is sealed into a SHA-256 chain that anyone with oversight can verify.
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Bid the prize down
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The lowest valid bid wins
A Z-score check holds any underbid that is statistically out of line.
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Everyone shares the discount
What the winner gives up is paid back to the group as dividends.
ermissioned groups · KYC-verified members · Escrow held until fraud checks clear
Sign in
Use the email and password you registered with.
Email
oversight@chainchit.io
Password
Password@123
Invalid email or password
Sign in
New to ChainChit? Create a member account
Trying the seeded demo?
Fill in the foreman login


next-server (v16.4.3)
Windows PowerShell
PS C:\Users\pakhi\Downloads\chainchit-main> npm run typecheck
> chainchit@0.1.0 typecheck
> tsc --noEmit
PS C:\Users\pakhi\Downloads\chainchit-main> npm test
> chainchit@0.1.0 test
> node --experimental-strip-types --test tests/fraud-engine.test.ts
(node:26384) [MODULE_TYPELESS_PACKAGE_JSON] Warning: Module type of file:///C:/Users/pakhi/Downloads/chainchit-main/tests/fraud-engine.test.ts is not specified and it doesn't parse as CommonJS.
Reparsing as ES module because module syntax was detected. This incurs a performance overhead.
To eliminate this warning, add "type": "module" to C:\Users\pakhi\Downloads\chainchit-main\package.json.
(Use `node --trace-warnings ...` to show where the warning was created)
▶ Statistical Bid-Fraud Detection Mathematical Engine
✔FT-13: Normal competitive bid within window is accepted (1.3046ms)
✔FT-15: Abnormal deep underbid (₹60,000) is flagged as suspicious (0.2981ms)
✔FT-16: Winner selection correctly identifies lowest VALID bid (1.4652ms)
✔Statistical Bid-Fraud Detection Mathematical Engine (4.6791ms)
▶ Tamper-Evident SHA-256 Audit Log Hash Chaining (NFR13)
✔detects tampering when payload or previous hash is modified (0.8546ms)
✔Tamper-Evident SHA-256 Audit Log Hash Chaining (NFR13) (1.1286ms)
ℹtests 4
ℹsuites 2
ℹpass 4
ℹfail 0
ℹcancelled 0
ℹskipped 0
ℹtodo 0
ℹduration_ms 216.8395

```
PS C:\Users\pakhi\Downloads\chainchit-main> npm test

> chainchit@0.1.0 test
> node --experimental-strip-types --test tests/fraud-engine.test.ts

(node:26384) [MODULE_TYPELESS_PACKAGE_JSON] Warning: Module type of file:///C:/Users/pakhi/Downloads/chainchit-main/tests/fraud-engin
e.test.ts is not specified and it doesn't parse as CommonJS.
Reparsing as ES module because module syntax was detected. This incurs a performance overhead.
To eliminate this warning, add "type": "module" to C:\Users\pakhi\Downloads\chainchit-main\package.json.
(Use `node --trace-warnings ...` to show where the warning was created)
▶ Statistical Bid-Fraud Detection Mathematical Engine
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▶ Tamper-Evident SHA-256 Audit Log Hash Chaining (NFR13)
  ✔detects tampering when payload or previous hash is modified (0.8546ms)
✔Tamper-Evident SHA-256 Audit Log Hash Chaining (NFR13) (1.1286ms)
ℹtests 4
ℹsuites 2
ℹpass 4
ℹfail 0
ℹcancelled 0
ℹskipped 0
ℹtodo 0
ℹduration_ms 216.8395
PS C:\Users\pakhi\Downloads\chainchit-main>    echo $env:SEED_PASSWORD
PS C:\Users\pakhi\Downloads\chainchit-main>    echo $env:SEED_PASSWORD
PS C:\Users\pakhi\Downloads\chainchit-main> Remove-Item Env:SEED_PASSWORD
```

```
├ ƒ /api/dashboard
├ ƒ /api/disputes
├ ƒ /api/disputes/[id]
├ ƒ /api/escrow/withdraw
├ ƒ /api/exports/verify
├ ƒ /api/foreman/cycles
├ ƒ /api/foreman/cycles/[id]
├ ƒ /api/foreman/groups
├ ƒ /api/foreman/groups/[id]
├ ƒ /api/foreman/groups/[id]/members
├ ƒ /api/foreman/kyc
├ ƒ /api/foreman/kyc/[userId]/documents/[docId]
├ ƒ /api/foreman/users
├ ƒ /api/fraud/override
├ ƒ /api/groups/[id]/export
├ ƒ /api/groups/[id]/ledger
├ ƒ /api/kyc
├ ƒ /api/ledger/public-key
├ ƒ /api/maintenance
├ ƒ /api/maintenance/sweep
├ ƒ /api/notifications
├ ƒ /api/receipts/[id]
├ ○ /audit
├ ƒ /cycle/[id]
├ ○ /dashboard
├ ○ /disputes
├ ƒ /disputes/[id]
├ ○ /foreman
├ ○ /groups
├ ƒ /groups/[id]
├ ○ /kyc
├ ○ /login
├ ○ /notifications
├ ƒ /receipts/[id]
└ ○ /register


ƒ Proxy (Middleware)

○  (Static)   prerendered as static content
ƒ  (Dynamic)  server-rendered on demand

PS C:\Users\pakhi\Downloads\chainchit-main> npm start

> chainchit@0.1.0 start
> next start

▲ Next.js 16.4.3
- Local:         http://localhost:3000
- Network:       http://172.28.16.1:3000
✓ Ready in 281ms
✓ Running next.config.ts took 68ms
```

```
PS C:\Users\pakhi\Downloads\chainchit-main> npm run build

> chainchit@0.1.0 build
> next build --webpack

▲ Next.js 16.4.3 (webpack)
✓ Running next.config.ts took 103ms

  Creating an optimized production build ...
✓ Compiled successfully in 24.5s
  Running TypeScript ...
  Finished TypeScript in 8.7s ...
  Collecting page data using 7 workers ...
✓ Collecting page data using 7 workers in 2.3s
  Generating static pages using 7 workers (0/35) ...
  Generating static pages using 7 workers (8/35)
  Generating static pages using 7 workers (17/35)
  Generating static pages using 7 workers (26/35)
✓ Generating static pages using 7 workers (35/35) in 1256ms
  Finalizing page optimization ...
  Collecting build traces ...
✓ Collecting build traces in 9.4s
✓ Finalizing page optimization in 9.4s

Route (app)
┌ ○ /
├ ○ /_not-found
├ ƒ /api/audit
├ ƒ /api/audit/export
├ ƒ /api/auth/login
├ ƒ /api/auth/logout
├ ƒ /api/auth/register
├ ƒ /api/auth/session
├ ƒ /api/contributions
├ ƒ /api/cycles
├ ƒ /api/cycles/[id]
├ ƒ /api/cycles/[id]/bid
├ ƒ /api/cycles/[id]/close
├ ƒ /api/cycles/[id]/settle
├ ƒ /api/cycles/[id]/simulate
├ ƒ /api/cycles/[id]/stream
├ ƒ /api/dashboard
├ ƒ /api/disputes
├ ƒ /api/disputes/[id]
├ ƒ /api/escrow/withdraw
├ ƒ /api/exports/verify
├ ƒ /api/foreman/cycles
├ ƒ /api/foreman/cycles/[id]
├ ƒ /api/foreman/groups
├ ƒ /api/foreman/groups/[id]
├ ƒ /api/foreman/groups/[id]/members
├ ƒ /api/foreman/kyc
```

```
next-server (v16.4.3)   Windows PowerShell
PS C:\Users\pakhi\Downloads\chainchit-main> npm run db:reset

> chainchit@0.1.0 db:reset
> tsx scripts/db-setup.ts --reset && tsx lib/db/seed.ts

schema reset
No config path provided, using default 'drizzle.config.ts'
Reading config file 'C:\Users\pakhi\Downloads\chainchit-main\drizzle.config.ts'
Using 'pg' driver for database querying
[✓] Pulling schema from database...
[✓] Changes applied
triggers installed
Seeded: foreman@chainchit.io and oversight@chainchit.io (password from SEED_PASSWORD, default Password@123).
Members register themselves via /register.
PS C:\Users\pakhi\Downloads\chainchit-main> npm run dev

> chainchit@0.1.0 dev
> next dev --webpack

▲ Next.js 16.4.3 (webpack)
- Local:         http://localhost:3000
- Network:       http://172.28.16.1:3000
✓ Ready in 1097ms
✓ Running next.config.ts took 70ms

○ Compiling /api/auth/login/route ...
 GET /api/auth/session 200 in 4.8s (next.js: 4.1s, proxy.ts: 295ms, application-code: 387ms)
 POST /api/auth/login 200 in 5.3s (next.js: 4.1s, proxy.ts: 299ms, application-code: 880ms)
 GET /api/auth/session 200 in 49ms (next.js: 6ms, proxy.ts: 28ms, application-code: 16ms)
○ Compiling /(app)/audit/page ...
 GET /audit 200 in 6.5s (next.js: 6.3s, proxy.ts: 15ms, application-code: 214ms)
 GET /login 200 in 3.1s (next.js: 1613ms, proxy.ts: 456ms, application-code: 1079ms)
 GET /audit 200 in 2.7s (next.js: 878ms, proxy.ts: 1629ms, application-code: 202ms)
 GET /api/auth/session 200 in 3.2s (next.js: 1613ms, proxy.ts: 456ms, application-code: 1126ms)
 GET /api/auth/session 200 in 324ms (next.js: 9ms, proxy.ts: 153ms, application-code: 162ms)
 GET /api/auth/session 200 in 79ms (next.js: 15ms, proxy.ts: 21ms, application-code: 44ms)
 GET /api/auth/session 200 in 90ms (next.js: 35ms, proxy.ts: 21ms, application-code: 33ms)
 GET /audit 200 in 136ms (next.js: 11ms, proxy.ts: 13ms, application-code: 112ms)
 GET /audit 200 in 103ms (next.js: 21ms, proxy.ts: 21ms, application-code: 61ms)
 GET /api/cycles 200 in 2.2s (next.js: 1860ms, proxy.ts: 37ms, application-code: 320ms)
 GET /api/cycles 200 in 2.2s (next.js: 1860ms, proxy.ts: 37ms, application-code: 351ms)
 GET /api/notifications 200 in 2.3s (next.js: 1859ms, proxy.ts: 37ms, application-code: 409ms)
 GET /api/audit 200 in 2.4s (next.js: 1860ms, proxy.ts: 37ms, application-code: 548ms)
 GET /api/notifications 200 in 2.7s (next.js: 1860ms, proxy.ts: 37ms, application-code: 837ms)
 GET /api/audit 200 in 2.7s (next.js: 1860ms, proxy.ts: 37ms, application-code: 849ms)
 GET /api/cycles 200 in 39ms (next.js: 7ms, proxy.ts: 15ms, application-code: 17ms)
 GET /dashboard 200 in 2.4s (next.js: 2.3s, proxy.ts: 7ms, application-code: 80ms)
 GET /groups 200 in 1395ms (next.js: 1315ms, proxy.ts: 15ms, application-code: 65ms)
 GET /api/notifications 200 in 328ms (next.js: 16ms, proxy.ts: 20ms, application-code: 292ms)
 GET /dashboard 200 in 338ms (next.js: 56ms, proxy.ts: 201ms, application-code: 80ms)
 GET /api/foreman/groups 200 in 2.5s (next.js: 2.1s, proxy.ts: 20ms, application-code: 356ms)
 GET /api/foreman/groups 200 in 2.5s (next.js: 2.1s, proxy.ts: 20ms, application-code: 369ms)
 GET /api/notifications 200 in 148ms (next.js: 14ms, proxy.ts: 19ms, application-code: 114ms)
```